

Network Cable and Connectors

Q1. Cat6 Ethernet cable: T568B colour order and RJ-45 pins

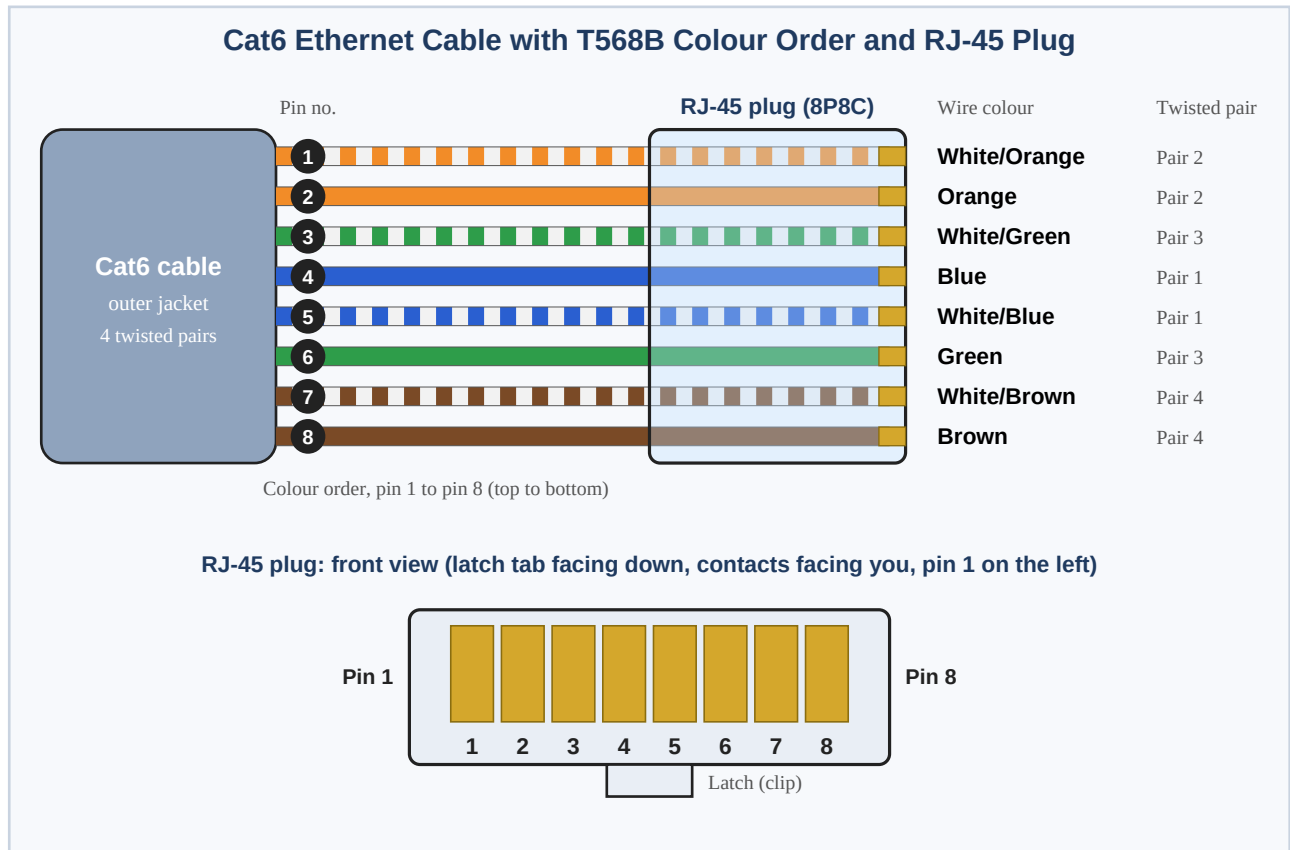


Figure 1: T568B wiring of a Cat6 cable and the pin numbers of an RJ-45 connector

Pin	Wire colour (T568B)	Pair	Function (10/100 Mbps)
1	White/Orange	Pair 2	TX+
2	Orange	Pair 2	TX-
3	White/Green	Pair 3	RX+
4	Blue	Pair 1	Unused (used in Gigabit)
5	White/Blue	Pair 1	Unused (used in Gigabit)
6	Green	Pair 3	RX-
7	White/Brown	Pair 4	Unused (used in Gigabit)
8	Brown	Pair 4	Unused (used in Gigabit)

Colour order to remember: White/Orange, Orange, White/Green, Blue, White/Blue, Green, White/Brown, Brown. Gigabit and faster speeds use all 8 wires.

Q2. Three types of network cables and where they are used

Cable	Real-world use	Why
Cat5e Up to 1 Gbps, 100 m	Cafe or restaurant POS terminal (like a Zomato POS), CCTV camera	A POS terminal only needs a steady, moderate-speed link over a short run. Cat5e is cheap and 1 Gbps is more than enough.
Cat6 1 Gbps up to 100 m; 10 Gbps up to about 55 m	WiFi router to gaming PC, console or smart TV	Games and streaming need fast, low-latency links. Cat6 has better shielding from crosstalk, so it is more stable and faster than Cat5e.
Fibre optic 10 Gbps and higher, kilometres	ISP lines (Jio/Airtel fibre) and the data-centre links of an IPL streaming server	Data travels as light, so it carries huge traffic over long distances with no electrical interference and very little signal loss.

Q3. How to crimp an RJ-45 connector onto a network cable

Step	What to do
1	Keep the tools ready: Cat6 cable, RJ-45 plugs, crimping tool, and a cable stripper or cutter.
2	Cut the cable end straight, then remove about 2 to 3 cm of the outer jacket without nicking the wires inside.
3	Untwist the four pairs, cut off any plastic spline or pull string, and straighten each wire with your fingers.
4	Arrange the wires in T568B order: White/Orange, Orange, White/Green, Blue, White/Blue, Green, White/Brown, Brown.
5	Hold them flat and side by side, then trim the ends evenly so about 1.2 cm of wire sticks out from the jacket.
6	Hold the plug with the latch facing down and slide the wires in until every wire touches the end of the plug. The jacket should go a little inside too.
7	Put the plug in the crimping tool and squeeze firmly until it clicks. This pushes the gold pins into the wires.
8	Repeat on the other end with the same T568B order, then test the cable with a cable tester.

Two common mistakes to avoid:

1. **Wrong colour order or different standards on the two ends.** The cable may not work or may run slowly. Both ends should be T568B (or both T568A) for a normal cable.
2. **Wires not pushed fully in or untwisted too much.** The pins then miss the copper, giving loose contact, and too much untwisting causes crosstalk and slow speed. Leave the wires twisted as close to the plug as possible.

Q4. Gaming zone for an IPL screening event

For many screens and consoles, wired connections are best because WiFi gets crowded and unstable when many people and devices are in one hall. I would use the following:

Connection	Cable and connector	Why
Router/switch to each console, PC or streaming box	Cat6 (or Cat6a) with RJ-45	Gigabit speed, low latency and stable, and cheaper than fibre for short runs up to 100 m.
Main switch to switches at other corners	Cat6a with RJ-45 for short runs, or fibre optic (multimode) with LC connectors for long runs	Carries the combined traffic of many devices at 10 Gbps with no interference from power cables and equipment.
ISP line into the venue	Fibre optic with SC/LC connector	The live IPL stream needs high, steady bandwidth and fibre gives reliable speed.
Console or streaming box to screen	HDMI cable with HDMI connector	Carries 4K video and audio to the screens with no lag.

Extra tips: use a gigabit switch with enough ports, label every cable, and use short, neat patch cables so nobody trips on them and connectors are not damaged.

Q5. Network cable in use: behind a WiFi router

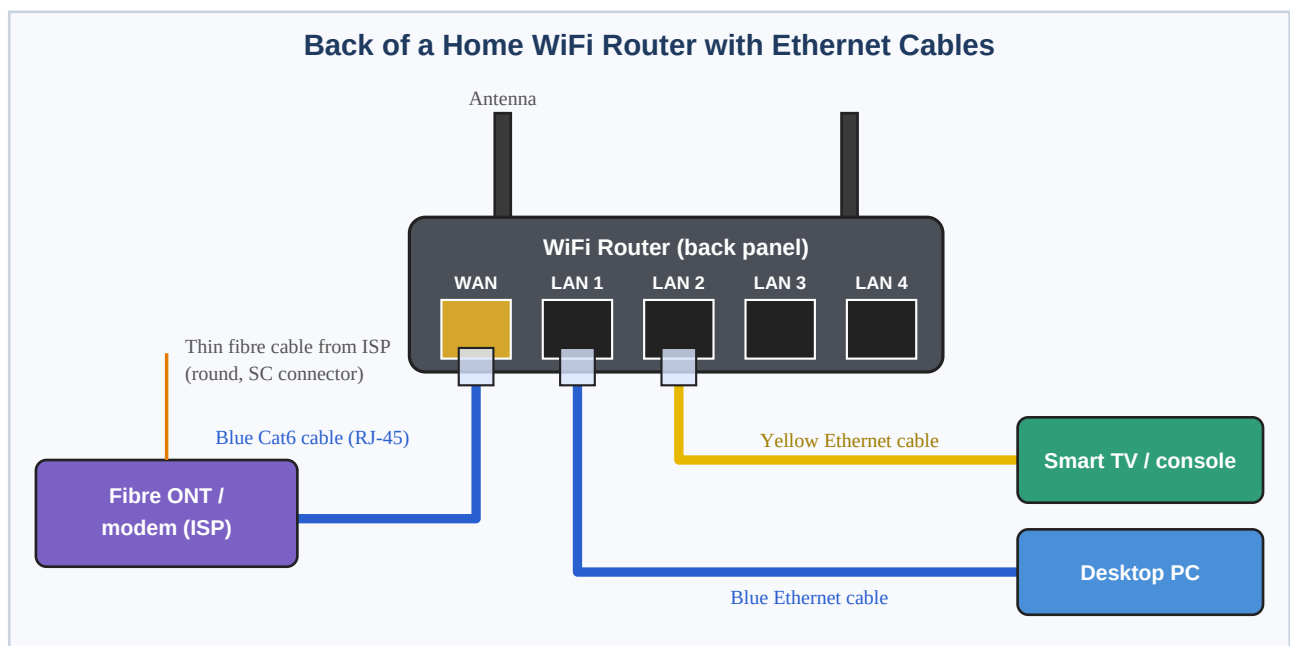


Figure 2: Illustration of Ethernet cables (clear plastic RJ-45 plugs) in the back of a home WiFi router

Type: The blue and yellow cables are Cat6 (or Cat5e) Ethernet patch cables with clear plastic RJ-45 connectors. The thin round cable from the ISP is fibre optic with an SC connector.

What they connect: One blue cable joins the fibre ONT/modem to the router's WAN port and brings the Internet in. The cables in the LAN ports connect the router to a desktop PC and to a smart TV or gaming console.